

dielectric elastomers, inorganic compounds, ceramics, nanomaterials, and composites. These materials impart flexibility and durability, supporting various sensor applications. Specific components like carbon-based inks, electroactive polymers, and conductive metallic inks or carbon nanotubes fulfil distinct functions in pressure, strain, and touch sensors, while innovations like quantum dots in photodetectors broaden their capabilities.

These sensors are designed to be compatible with flexible substrates and easily conform to curved surfaces. Scalable manufacturing enables the production of large-scale, multi-functional layers, which are vital for creating compact, integrated devices. Despite their versatility and essential roles across automotive, healthcare, and consumer electronics sectors, high costs and technical complexities restrict their wider industrial adoption.

Economic puzzle: Balancing revenue and costs

Printed sensors offer groundbreaking potential by integrating sensing capabilities onto flexible, large-scale substrates through advanced printing techniques. However, their market revenue remains limited, representing just a fraction of the multi-billion dollar global sensor market, indicating untapped opportunities.

Printed sensors present a dual narrative in technological innovation, offering revolutionary functionalities but facing challenges due to high production costs and competition from traditional, lower-cost sensor technologies. Their success relies on cost reduction, production scaling, and niche establishment. While their revenue potential may grow with maturity and expansion into more applications, they currently struggle to compete with established, cost-effective sensor technologies.

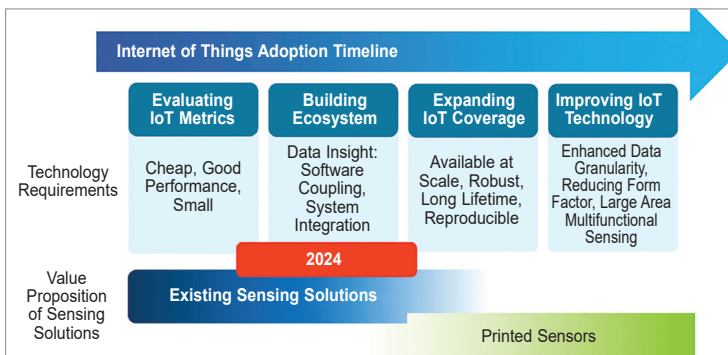
Market research attributes the disparity in market capture of printed sensors to their high production costs

Impact of early adoption challenges

From inception to broad use, the adoption timeline for printed sensors showcases technology progress and market readiness, highlighting a shift from niche to mainstream applications in sectors like IoT.

- Traditional sensors are initially preferred in early IoT deployments for their proven reliability and cost-effectiveness. However, as demands evolve, the unique features of printed sensors—such as flexibility and multifunctionality—become increasingly vital, particularly for complex tasks like smart inventory management.
- Printed sensors also enhance AI and AR technologies. For example, rubber sensors in AR gloves deliver precise hand movement data, enriching user interactions.

Despite their advantages, the widespread adoption of printed sensors faces obstacles primarily due to the market's preference for established, cost-effective solutions and the ongoing need to demonstrate their reliability and cost-effectiveness.



and niche applications. Unlike traditional sensors, printed sensors have not achieved widespread adoption that could drive down costs through mass production, resulting in relatively limited revenue generation compared to the overall sensor market.

The production process of printed sensors involves specialised ink formulation, printing, and curing, distinct from the subtractive manufacturing used for traditional sensors. These steps contribute to higher initial costs due to the need for bespoke materials and formulations that aren't yet commoditised. For instance, while some printed pressure sensors use conductive carbon inks and are relatively inexpensive, emerging materials for other sensor types, such as photodetectors and capacitive touch sensors, remain costly due to their specialised nature and limited production scale.

Manufacturing advancements and navigating technical hurdles

Printed sensors are emerging as a transformative technology in the ever-expanding Internet of Things (IoT).

These sensors seamlessly integrate with edge computing and AI to enhance device functionality, offering expansive coverage for applications such as autonomous driving. Their flexibility allows for integration into complex structures, and their cost-effectiveness supports scalable deployments.

Printed sensors are increasingly vital in the automotive industry. They enhance monitoring systems through edge computing for real-time data processing. This is particularly important for managing the battery health of electric vehicles, enhancing safety, and preventing failures. While mass production of these sensors is currently feasible, advancements in manufacturing, such as roll-to-roll and 3D printing, could improve scalability and reduce costs.

Incorporating printed sensors into IoT systems poses challenges, including competition with traditional sensors, but they offer substantial benefits in industrial IoT applications like predictive maintenance, asset tracking, and process optimisation. For example, printed piezoelectric